Zonghe Shao

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Education

Huazhong University of Science & Technology(HUST) *Sep. 2020-Jul. 2024*
B.Eng. in ***Artificial Intelligence***

- Overall GPA: 3.93/4.0 (91.72/100) Ranking: 2/26
- National Encouragement Scholarship, Ministry of Education of PRC (**Top** 3% Natinal Wide), **TWICE**

Research Interests

GOAL: I hope to build a perceptive, interpretable and general artificial intelligence.
Machine Learning & Computer Vision

- I work on the application of computer vision to medical treatment to assist physicians in diagnosis. Particularly, I focus on using vision to analyze facial expressions and body movements, as well as medical image processing
- I focus on extracting depth information from images and reconstructing the 3D world. Specifically, I would like to build algorithm into cameras or apply it in autonomous driving
- I did some work on the analysis and prediction of spatio-temporal series data, and I am looking at completing it with a more streamlined algorithm now
- I am also interested in optimization theory and non-convex optimization in machine learning

Efforts towards the goals

- Joined the HAIV lab and deep reinforcement learning lab and took the initiative to learn the basics and cutting edge work in the field
- Pondered the problems the current work faced and put forward my views, and actively discussed my views with others
- Proactively verified the feasibility of my ideas and tried to make my innovative work on AI algorithms

Publications

A Novel Data-Driven LSTM-SAF Model for Power System Transient Stability Assessment
Zonghe Shao,Qichao Wang,Yuzhe Cao,Defu Cai,Renzhi Lu
In Submission

Research Experience

HAIV Lab, HUST Wuhan, China
Advisor: Assistant Prof. Xiang Xiang *Dec.2022-Present*
Topic: Patient Pain Intensity Estimation Based on Face in Complex Scenarios

- Studied the robust representation of human face based on action units and learned the method for recognizing asymmetric faces in harsh environments
- created more general dataset based on existing datasets such as UNTSC by introducing various harsh environments such as outdoor and indoor low light.
- Todo: benchmark the intensity of action units, propose facial treatment strategy to improve model robustness

Deep Reinforcement Learning Lab, HUST Wuhan, China
Advisor: Assistant Prof. Renzhi Lu *Feb.2022-Jan.2023*
Topic: Machine Learning on Small Datasets with Long-tailed Distributions

- Proposed efficient feature extraction method based on power angle clusters and improved wrapper for power system stability assessment with variable topology
- Redesigned the focal loss function and made it change the parameters to focus on a small number of samples during training
- Constructed an LSTM-SAF method that adequately mine the information albeit in small datasets and gave validation in multiple datasets

Project Experience

Artificial intelligence elite training program in Baidu *Nov.2022-Present*
Remote Member

- Learned and mastered the design and application of AI algorithms based on PaddlePaddle in Baidu
- Provided standard solutions to open algorithm design problems on the platform matiji
- Designed the format and content of the post-class assignments for some chapters in the public machine learning course of Baidu ai studio

Design of the software named "Ping An Good Car Owner" *Dec. 2020-May 2021*
Team Leader

- Constructeed a human-computer interaction software based on C language
- Realized a variety of software functions, such as vehicle registration, insurance purchase and claim settlement
- Constructed a data management method based on the dual mode of user and administrator

Extra-curriculum Activity

Alumni Association and Student Development Guidance Center *Oct.2021-Oct.2022*
Member

- Planned and organized several alumni homecoming lectures and acted as the venue director
- Arranged and coordinated the schedule of returning alumni in the 70th anniversary of the university and acted as the area manager in the main venue
- Hosted the Light Run on campus with over 10,000 participants

Grants Committee *Oct.2020-Sep.2022*
Directors

- Manage and review students' qualifications for financial aid and scholarships
- Hosted public campaign speeches for various scholarships and honors
- Led the Grants Committee to campus team of the year honors during my tenure as head

Volunteer Association *Sep.2020-Present*
Member

- Persisted in volunteering activities and was determined to contribute my efforts in environmental protection and social service.
- Served as a session leader and assisted the moderator in organizing the session agenda at the Artificial Intelligence Conference in China Optics Valley

Awards and Honors

Scholarship of National Encouragement,TWICE *Sep.2022 & Sep.2021*
Scholarship of Academic Excellence,TWICE *Sep.2022 & Sep.2021*
Meritorious Winner, Mathematical Contest in Modeling (MCM) *Feb.2022*
National Second Prize, The Chinese Mathematics Competition (CMC) *Nov.2021*
Scholarship of Outstanding Freshman *Jan.2021*

Skills and Interests

- **Languages**: Chinese(native), English (fluent, **academic writing**)
- **Programming**: C/C++, Python, MATLAB
- **DL Framework**: Pytorch, Tensorflow
- **Amateur Interests**: table tennis, badminton, yoga